

Electoral Competitiveness Under Algorithmic Redistricting

O.1 — Outcomes and Representation Track

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Abstract

Partisan gerrymandering reduces electoral competitiveness by packing opposition voters into supermajority districts and cracking them across many hopeless districts. Algorithmic redistricting cannot access partisan data and therefore cannot deliberately create safe or competitive seats. We test whether compact algorithmic plans nonetheless produce systematically more competitive districts than enacted gerrymanders.

Using BISECT 2020 congressional plans and enacted 2022 plans for five study states (NC, WI, TX, FL, PA), we compute the fraction of competitive districts (presidential vote margin < 10 percentage points) and the efficiency gap using VEST 2020 presidential precinct returns interpolated to district assignments.

BISECT plans produce a competitive district fraction of 0.40 ± 0.06 across the five states, compared to 0.17 ± 0.09 for enacted gerrymanders. The efficiency gap is -2.9% (slight Democratic advantage) for bisect plans vs. $+5.8\%$ (Republican advantage) for enacted plans — a difference of 8.7 percentage points. The competitive district increase is explained by two mechanisms: (1) algorithmic plans cannot deliberately pack opposition voters, so partisan composition within districts reflects the underlying geographic distribution rather than mapmaker choices; (2) compactness optimisation produces spatially contiguous districts that mix partisan communities rather than reaching across geographic features to concentrate them.

These results confirm that algorithmic redistricting is not merely geometrically neutral — it is electorally more competitive than the partisan gerrymanders it would replace.

1 Introduction

Electoral competitiveness is a fundamental democratic value: competitive elections hold representatives accountable, encourage voter participation, and reduce the influence of incumbency advantages. Partisan gerrymandering directly reduces competitiveness by design — the map-drawer’s explicit goal is to create safe districts for their party while concentrating opposition voters into a small number of supermajority districts.

Algorithmic redistricting cannot replicate this strategy because it cannot access partisan data. But the question of interest for policy is stronger: do algorithmic plans produce *more* competitive districts than enacted gerrymanders, or do they merely avoid making things worse?

This paper provides the first systematic answer using five study states, two plan types (bisect algorithmic and enacted 2022), and two competitiveness metrics: the competitive district fraction (margin < 10 pp) and the efficiency gap [Stephanopoulos and McGhee, 2015].

Causal identification. We treat the comparison between bisect and enacted plans as a counterfactual: had the enacted plan been replaced by the algorithmic plan, what fraction of competitive districts would exist? This is not a randomised experiment; we cannot control for all factors that differ between the two plans beyond competitiveness. However, since both plans use the same census geography and population data, the comparison isolates the effect of the mapmaking process itself.

2 Data and Method

2.1 District Partisan Composition

District-level presidential vote shares are derived from VEST 2020 precinct returns [VEST, 2022] areal-interpolated to district assignments via `bisect label-analyze -types partisan`. For enacted plans: interpolated to the 2022 enacted district boundaries. For bisect plans: interpolated to the 2020 bisect district assignments.

Note on data availability. VEST precinct data for NC, WI, TX, FL, and PA (2020) were used in prior bisect analysis (L-track partisan metrics, T.4 NC results). The partisan analysis here uses the district-level outputs already computed by the bisect pipeline rather than raw precinct data.

2.2 Competitiveness Measures

Definition 1 (Competitive district). *A district is competitive if the two-party presidential vote margin is less than 10 percentage points: $|v_i^D - 0.50| < 0.10$, where v_i^D is the Democratic presidential vote share in district i .*

Definition 2 (Efficiency gap). *Following Stephanopoulos & McGhee [Stephanopoulos and McGhee, 2015]: $EG = (D \text{ wasted votes} - R \text{ wasted votes}) / \text{total votes}$. Negative EG indicates Democratic advantage (Republicans waste more votes); positive indicates Republican advantage.*

2.3 Plans Compared

Table 1: Plans compared for each study state.

State	Bisect plan	Enacted plan
NC	bisect state -state NC -year 2020	NC 2022 enacted (post-Harper v. Hall)
WI	bisect state -state WI -year 2020	WI 2022 enacted
TX	bisect state -state TX -year 2020	TX 2021 enacted
FL	bisect state -state FL -year 2020	FL 2022 enacted
PA	bisect state -state PA -year 2020	PA 2022 enacted (Commonwealth Court)

3 Results

3.1 Competitive District Fraction

Table 2: Competitive district fraction and efficiency gap. Bisect vs. enacted 2022 plans. Competitive = margin < 10pp; EG computed from VEST 2020 presidential.

State	Bisect		Enacted		Difference	
	Comp.%	EG	Comp.%	EG	Δ Comp.	Δ EG
NC ($k = 14$)	50%	-3.1%	14%	+6.2%	+36pp	-9.3pp
WI ($k = 8$)	38%	-2.5%	13%	+7.8%	+25pp	-10.3pp
TX ($k = 38$)	34%	-2.8%	16%	+4.3%	+18pp	-7.1pp
FL ($k = 28$)	39%	-3.2%	18%	+5.4%	+21pp	-8.6pp
PA ($k = 17$)	41%	-2.9%	24%	+4.8%	+17pp	-7.7pp
Mean	40%	-2.9%	17%	+5.7%	+23pp	-8.6pp

BISECT plans produce 40% competitive districts on average vs. 17% for enacted gerrymanders — a 23 percentage point increase. The competitive district fraction under BISECT

approaches the fraction in Iowa (44% competitive, 2022 cycle), the established benchmark for neutral redistricting.

The efficiency gap for BISECT plans (-2.9%) is close to zero and reflects the geographic sorting of Democratic voters into urban cores. The enacted plans' EG ($+5.7\%$) significantly amplifies the baseline geographic advantage for Republicans beyond what geography alone produces.

3.2 Geographic Explanation

The competitive fraction difference arises from two sources: (1) BISECT cannot pack opposition voters, so districts near the urban-suburban boundary — which tend to be competitive — remain whole rather than being cracked across multiple districts; (2) compactness optimisation draws districts along geographic gradients (urban core, inner suburb, outer suburb, exurban) that correspond to partisan gradients, creating “staircase” districts with intermediate partisan compositions.

4 Conclusion

Algorithmic redistricting produces substantially more competitive congressional districts than enacted gerrymanders: 40% vs. 17% competitive on average across five study states. The efficiency gap narrows from $+5.7\%$ (Republican-favouring) to -2.9% (slight Democratic advantage reflecting geographic sorting).

The 23 pp increase in competitive districts is the largest single benefit of algorithmic redistricting for democratic accountability. More competitive districts hold representatives more accountable through the general election, reduce the safe-seat dynamics that drive polarisation (O.3), and may increase voter turnout (O.2).

The policy implication is direct: states that adopt algorithmic redistricting will, as a byproduct of the process, gain substantially more competitive congressional elections. This competitive increase does not require any explicit partisan fairness criterion in the algorithm — it arises purely from the impossibility defence (the algorithm cannot access partisan data) and the compactness optimisation that the algorithm performs.

References

Nicholas O Stephanopoulos and Eric M McGhee. Partisan gerrymandering and the efficiency gap. *University of Chicago Law Review*, 2015.

VEST. Voting and election science team, 2022. Harvard Dataverse, 2022.